AI DEVELOPER (RAG)

Retrieval-Augmented Generation (RAG): Architecture & Engineering Guide

1. Why RAG for Enterprise AI?

Standard LLMs suffer from knowledge cutoffs and hallucinations on private domain data. RAG combines vector retrieval with generative synthesis to deliver grounded, accurate, and source-attributed responses without costly model fine-tuning.

2. Core RAG Pipeline Stages

Stage	Component	Best Practice / Optimal Parameter
Ingestion & Chunking	RecursiveCharacterTextSplitter	Chunk size: 500 chars, Chunk overlap: 50 chars
Vector Embedding	OpenAI text-embedding-3-small	1536 dimensions, normalized cosine distance
Vector Storage & Indexing	ChromaDB / Pinecone	HNSW index with cosine similarity metric
Context Retrieval	Similarity & MMR Search	Top-K = 3 to 5 chunks, similarity score threshold > 0.70
Synthesis & Prompting	GPT-3.5-Turbo / GPT-4	Strict grounding prompt with source citation requirement